Nguyen Hoang Nguyen

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Experience

AI Engineer *Ho Chi Minh City, Vietnam*
FPT Software AI Center *Dec 2023 – Present*

- Built a RAG indexing pipeline on Apache Airflow processing 500+ multi-format documents, using VLMs and TableFormer to extract nested tables and diagrams from multilingual PDFs, Excel, and PPTX, reaching **80% context recall**.
- Researched and shipped agentic systems on LangGraph – dynamic tool calling, planning, and multi-step reasoning for autonomous query resolution, plus a document-generation service that cut manual documentation effort by **22%**.
- Automated legacy-code understanding across 1M+ lines and 7 languages on 4 mainframe systems: a Gzip-compression + k-NN classifier at **90% accuracy**, and custom ANTLR4 parsers reaching **88% COBOL syntax coverage**.

Skills

3D Vision & Simulation: 3D & 2D Gaussian Splatting, gsplat, Neural Harmonic Textures, 3DGUT, COLMAP structure-from-motion, camera calibration, MCMC densification, Mask2Former, RoMa, Depth Anything, Rerun, NVIDIA Isaac Sim, Genesis

ML Engineering & LLMs: Python, PyTorch, OpenCV, NumPy, VLMs, LoRA & QLoRA, LangChain, LangGraph, Agentic RAG, Scikit-learn, Weights & Biases, pytest

Infrastructure & Tools: Modal, Docker, Linux, Git, FastAPI, Azure, Apache Airflow, PostgreSQL, MongoDB, Milvus, Neo4j, Redis, ANTLR4

Projects

Saigon Digital Twin: Photorealistic 3D City Streets from Public Imagery

- Building a walkable, photorealistic 3D model of downtown Ho Chi Minh City from public imagery only: **100+ street-level 360° panoramas** (1,200 views) covering a full street.
- Handled **transient objects** (cars, motorbikes, pedestrians) with **Mask2Former** and **Mask R-CNN**, plus a **multi-view consistency** check (**RoMa**) that keeps parked vehicles and removes moving traffic.
- Trained **2D Gaussian Splatting** from a dense **RoMa** point cloud, with sky and depth priors (**Depth Anything**, **COLMAP**) to handle reflective glass towers, sky haze and floating artifacts.
- Outperformed a standard 2DGS baseline on unseen street views with no floating sky Gaussians, renders **36% closer to real photos** (LPIPS from 0.40 to 0.26), PSNR from 18.4 to 19.3 dB.

3D Reconstruction of Drone-Captured Tower Sites, Reflective and See-Through Scenes *(VAI Novel-View-Synthesis Competition)*

- Reconstructed camera poses with **COLMAP** structure-from-motion, re-undistorted the input images with **Lanczos4** resampling, then trained **Neural Harmonic Textures** on **gsplat** through a **3DGUT** rasterizer, growing to 9M primitives with **MCMC densification** under a combined **L1 + D-SSIM + LPIPS** loss.
- Built an end-to-end pipeline turning photo sets into photorealistic, measurable 3D models, focusing on scenes where conventional reconstruction breaks down: a **thin lattice telecom tower** shot by drone over a crowded city block, which leaves almost no solid structure to lock onto; a **bonsai on a reflective glass table**, where the reflection reads as real geometry; and a **chair with a see-through woven mesh back**, achieving **PSNR 25.76, SSIM 0.852 and LPIPS 0.134** on independent novel-view evaluation.

Code as Policy *(Genesis, Franka Panda)*

- Used an **LLM as the robot's high-level controller**: plain-language instructions are converted into Python commands that drive a Franka Panda arm in the **Genesis** physics simulator, guided by an overhead **depth camera** with **open-vocabulary object detection (YOLOE)** that locates each object on the

table in 3D.

- Designed the **motion planning** for collision-free execution: **RRTConnect** plans the approach, and every grasp is validated before the arm moves.

Vietnamese-English Translation with Teencode and Slang Understanding

- Fine-tuned Vistral-7B (4-bit QLoRA) on a curated 8-task instruction dataset, achieving **34.9 BLEU** on VLSP2020 and beating GPT-3.5 and Google Translate on three Vietnamese social-media benchmarks by up to **+7.5 BLEU**. Accepted at EAI AICON.

Education

Ton Duc Thang University — *Ho Chi Minh City, Vietnam*
Bachelor of Science in Software Engineering — *Sep 2020 – Aug 2024*

- GPA 3.44/4.0. Undergraduate program taught in English.
- Member of the university's ICPC training team; core member of ICON Club (technical blogs, seminars, full-stack quiz platform).

Feng Chia University – Exchange Student, GPA 3.5/4.0 — *Taichung, Taiwan*, *Jul 2023 – Aug 2023*

Publications

Enhancing Vietnamese to English Translations With Slang and Teencode Understanding — Accepted

Nguyen Hoang Nguyen, Phuoc Tran, Dat Duong

EAI International Conference on Artificial Intelligence for Communications and Networks

Honors, Awards and Certificates

University Study Scholarship, 2020–2023 (academic performance) — *2023*

3rd place, Academic contest "Recursion" 2022 — *2022*

University Entrance Scholarship (national graduation exam scores) — *2020*

NLP Specialization; Generative AI with LLMs *(DeepLearning.AI)* — *2023*